

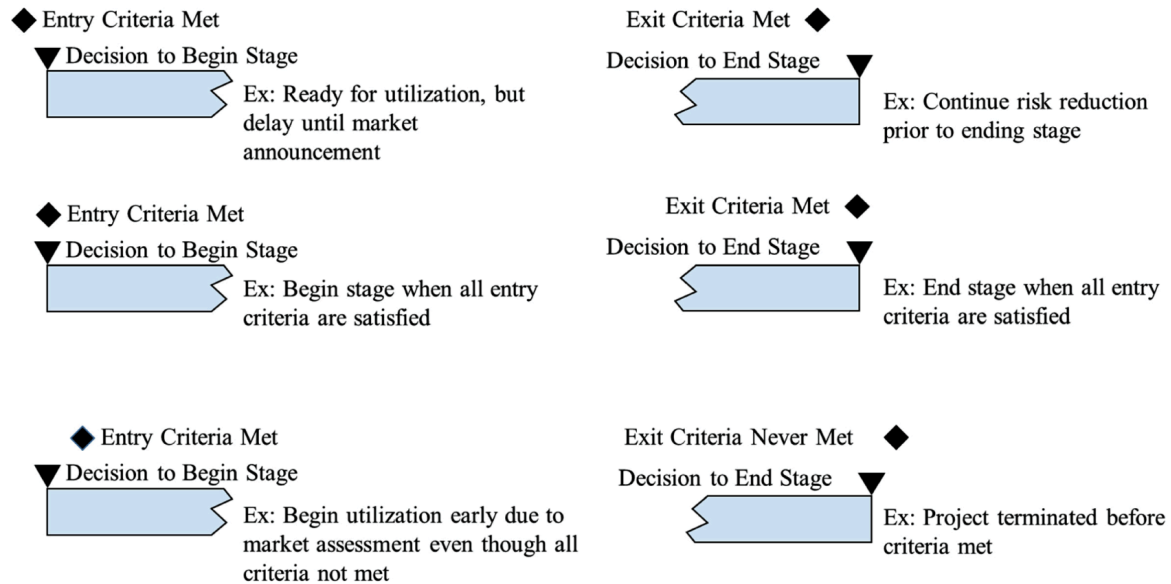


FIGURE 2.3 Criteria for decision gates. InCOSE SEH original figure created by yokell. Usage per the InCOSE notices page. All other rights reserved.

- the entry criteria are met, but the start of the stage is delayed;
- when the entry criteria are met, the decision to start the stage is made;
- although the entry criteria are not met, the stage is started;
- although the exit criteria are met, the decision to end the stage is delayed;
- when the exit criteria are met, the decision to end the stage is made;
- the decision to end the stage is made before the exit criteria are met.

At each decision gate, the options can include:

- Begin subsequent stage or stages;
- Continue this stage, possibly after some reformulation;
- Go to or restart a preceding stage;
- Hold the project activity;
- Terminate the project.

The option selected depends on the quality of the results of the effort so far (based on the answers to the exit questions) plus the risk of moving forward (based on the answers to the entrance questions). Stages do not need to be sequential. Transitions often occur, but it is common to have stages occurring concurrently. In complex systems, the decision can also be

more differentiated. For example: move part of the effort forward; hold on some; and terminate or reform others.

Decision gate approval follows review by qualified experts, involved stakeholders, and management. Approval should be based on evidence of compliance to the criteria of the review. Balancing the formality and frequency of decision gates is seen as a critical success factor for all SE process areas. The consequences of conducting a superficial review, omitting a critical discipline, or skipping a decision gate are usually long-term and costly.

It is important to note that there may be significant changes in the project's environment. This may impact the project's business case, system scope, or resources needed. Consequently, the related decision criteria should be updated and evaluated at every decision gate. Inadequate consideration can set up subsequent failures—usually a major factor in cost overruns and delays.

Upon successful completion of a decision gate, some artifacts (e.g., documents, analysis results, models, or other products of a system life cycle stage) will have been approved as the basis upon which future work must build. These artifacts are placed under configuration management along with the decisions made and the associated rationale and assumptions (see [Section 2.3.4.5](#)).

2.1.4 Technical Reviews and Audits

Technical reviews and audits are used to assess technical progress, coordinate activities, and determine the technical status of a system of interest (SoI). According to ISO/IEC 24748–8 / IEEE 15288.2 (2014):

A technical review is “a series of systems engineering activities conducted at logical transition points in a system life cycle, by which the progress of a [project] is assessed relative to its technical requirements using a mutually agreed-upon set of criteria” and

An audit is “a detailed review of processes, product definition information, documented verification of compliance with requirements, and an inspection of products to confirm that products have achieved their required attributes or conform to released product configuration definition information.”

The technical reviews and audits to be performed occur throughout the system life cycle and should be captured in the project's Systems Engineering Management Plan (SEMP) and reflected in the project's schedule (see [Section 2.3.4.1](#)). They may be part of a decision gate review (see [Section 2.1.3](#)). A representative set of technical reviews and audits are listed in [Table 2.1](#). They should be tailored for the needs of the project and the methodologies being used. ISO/IEC/IEEE 24748-1 (2018), Annex F and ISO/IEC 24748-8 / IEEE 15288.2 (2014) provide useful guidance for the planning and tailoring of reviews to the needs of the project and its stakeholders.

TABLE 2.1 Representative technical reviews and audits

Defense Projects per ISO/IEC/IEEE 24748-8/IEEE 15288.2 (2014)	Space Projects per NASA (2007b)	Incremental Commitment Spiral Model per Boehm, et al. (2014)
Alternative Systems Review (ASR)	Mission Concept Review (MCR)	Exploration Commitment Review
System Requirements Review (SRR)	System Requirements Review (SRR)	(ECR) Valuation
System Functional Review (SFR)	Mission Definition Review (MDR)	Commitment Review (VCR)
Preliminary Design Review (PDR)	System Definition Review (SDR)	Foundation Commitment Review
Critical Design Review (CDR)	Preliminary Design Review (PDR)	(FCR) Development
Test Readiness Review (TRR)	Critical Design Review (CDR)	Commitment Review _n (DCR _n)
Functional Configuration Audit (FCA)	System Integration Review (SIR)	Operations Commitment
System Verification Review (SVR)	Operational Readiness Review (ORR)	Review _n (OCR _n)
Production Readiness Review (PRR)	Flight Readiness Review (FRR)	
Physical Configuration Audit (PCA)	Mission Readiness Review (MRR)	
	Post-Launch Assessment Review (PLAR)	
	Critical Events Readiness Review (CERR)	

Defense Projects per ISO/IEC/IEEE 24748-8/IEEE 15288.2 (2014)	Space Projects per NASA (2007b)	Incremental Commitment Spiral Model per Boehm, et al. (2014)
	Post-Flight Assessment Review (PFAR)	
	Decommissioning Review (DR)	
	Disposal Readiness Review (DRR)	
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Figure 2.4 depicts the relationship between these reviews and audits identified in ISO/IEC 24748–8 / IEEE 15288–2 (2014) and the typical technical baselines across the system life cycle applicable for a sequential life cycle model. This depiction will vary significantly for incremental life cycle models.

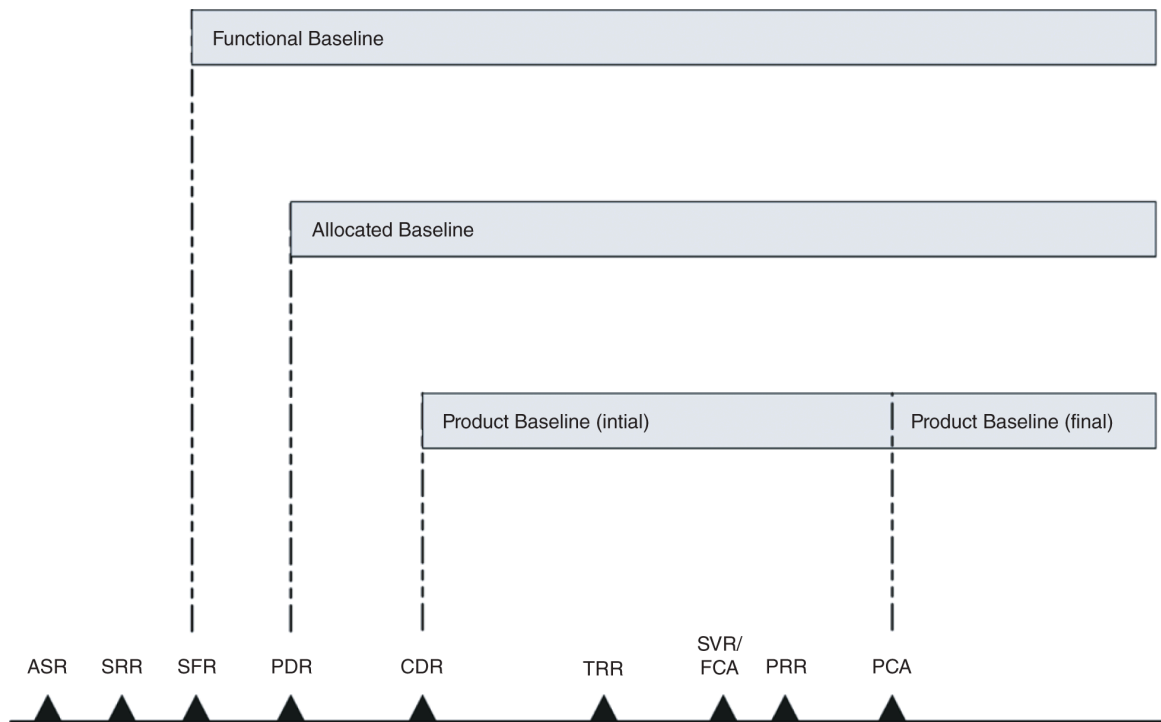


FIGURE 2.4 Relationship between technical reviews and audits and the technical baselines. From ISO/IEC 24748-8 / IEEE 15288.2 (2014). Used with permission. All other rights reserved.

Technical reviews and audits provide an opportunity to assess the following:

- The SoI is meeting its requirements
- The SoI is meeting stakeholder expectations, internal and external
- The SoI will have acceptable quality characteristics (QCs)
- The SoI is at an appropriate level of maturity
- The SoI is at an acceptable level of risk
- There is a clear path toward verifying and validating the SoI and its elements

Good practices for technical reviews and audits include:

- Plan the review or audit, including getting concurrence on a mutually agreeable location and date
- Application of multiple instances of the reviews and audits, both at multiple levels of the systems hierarchy and during each increment or iteration
- Elimination of unnecessary reviews or audits
- Establish clear entry and exit criteria for each review and audit

- Establish roles and responsibilities for the preparation, conduct, and acceptance of each review
- Make the reviews be risk-driven (risk is at an acceptable level) or event-driven (the entry criteria has been satisfied), not schedule-driven (must happen on a certain date)
- Consider “dry-runs” to make the review as efficient as possible
- Include subject matter experts (SMEs) and independent reviewers
- Include all members of the team, including acquirers and suppliers
- Capture clear actions, with ownership and due dates, for all issues that arise
- Follow up on actions that were raised

Each technical review or audit should include knowledgeable participants as well as participants with sufficient objectivity to assess satisfaction of the pre-established review criteria. Based on the purpose and level of the review, the participants may include representatives from the acquirer or supplier organizations, or both. A list of possible participants is provided below:

- Project Manager
- Lead SE Practitioner / Chief Engineer / Lead Engineer
- Review or Audit Chair
- Recorder (person charged with capturing the results of the review or audit)
- Acquirer Representative(s)
- Supplier Representative(s)
- Project Verification and Validation Lead
- Other Technical Leads

2.2 LIFE CYCLE MODEL APPROACHES

[Section 2.1](#) introduces the concept of life cycle stages. The life cycle models are thus the framework within which the individual life cycle stages and transitions between them are planned and implemented. There are many different life cycle models, each suitable for different situations. A common

way to differentiate them is to divide the life cycle model approaches into three groups: sequential, incremental, and evolutionary. [Figure 2.5](#) provides the general concept for each of these approaches, and [Table 2.2](#) summarizes their distinguishing characteristics.

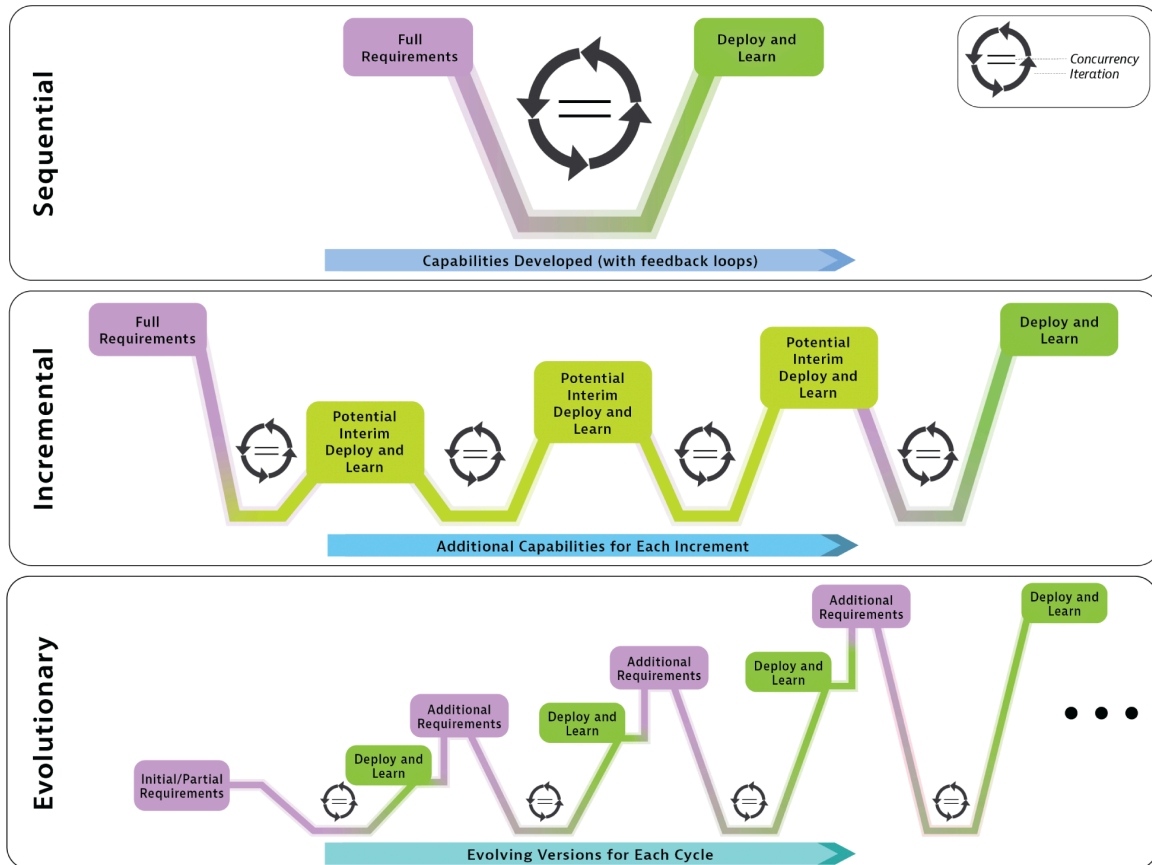


FIGURE 2.5 Concepts for the three life cycle model approaches. InCOSE SEH original figure created by Endler. Usage per the InCOSE nOTICES page. All other rights reserved.

TABLE 2.2 Life cycle model approach characteristics

Life cycle approach	Requirement set at start	Planned iterations	Multiple deployments
Sequential	Full	Single	No
Incremental	Full	Multiple	Potential
Evolutionary	Partial	Multiple	Typically

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ISO/IEC/IEEE 24748–1 (2018) provides further information on sequential (identified as “once-through”), incremental, and evolutionary life cycle model approaches. [Sections 2.2.1](#) to [2.2.3](#) elaborate on how these approaches can be applied to manage the work within each life cycle stage.

There are many factors that help determine which life cycle models are suitable for a specific system or project. Clause 6 of ISO/IEC/IEEE 24748–1 (2018) provides informational considerations that may influence the selection and adaptation of life cycle model, including:

1. stability of, and variety in, operational environments;
2. risks, commercial or performance, to the concern of stakeholders;
3. novelty, size, and complexity;
4. starting date and duration of utilization;
5. integrity issues such as safety, security, privacy, usability, availability;
6. emerging technology opportunities;
7. profile of budget and organizational resources available;
8. availability of the services of enabling systems;
9. roles, responsibilities, accountabilities, and authorities in the overall life cycle of the system;
10. the need to conform to other standards.

Other sources define characteristics and tailoring factors that can be used to guide tailoring. As an example, Project Management Institute (PMI) has published their Situation Context Framework (SCF) (2022) that “defines how to select and tailor a situation-dependent strategy for software development. The SCF is used to provide context for organizing your people, process, and tools for a software-based solution delivery team.” Seven dimensions (team size, geographic distribution, organizational distribution, skill availability, compliance, domain complexity, and solution complexity) with scaling factors in each dimension are defined within this framework.

As there is no universal approach, it is recommended that each organization continuously questions itself as to which approach, or

combination of approaches, is most suitable. Part IV of this handbook addresses tailoring and application considerations in more detail.

2.2.1 Sequential Methods

The sequential approach is focused on the general flow of the processes with feedback loops, but a single delivery. Sequential life cycle models break down SE activities into linear sequential stages, where each stage depends on the deliverables of the previous stages, along with feedback from subsequent stages.

On projects where it is necessary to coordinate large teams of people working in multiple companies, sequential approaches provide an underlying framework to provide discipline to the life cycle processes. Sequential life cycle models are characterized by a systematic approach that adheres to specified processes as the system moves through a series of representations from requirements through design to finished product. Specific attention is given to the completeness of documentation, traceability from requirements, and verification of each representation after the fact.

The strengths of sequential life cycle models are predictability, stability, repeatability, and high assurance. Process improvement focuses on increasing process capability through standardization, measurement, and control. These models rely on “master plans” to anchor their processes and provide project-wide communication. Historical data is usually carefully collected and maintained as inputs to future planning to make projections more accurate (Boehm and Turner, 2004).

The waterfall model, introduced by Royce (1970), was used to characterize the advantages and disadvantages of sequential approaches. The waterfall model has been used successfully in the manufacturing and construction industries, where the highly structured physical environments meant that design changes became prohibitively expensive much sooner in the development process. In addition, safety-critical products, such as the Therac-25 medical equipment (see the case study in [Section 6.1](#)), can only

meet modern certification standards by following a thorough, documented set of plans and specifications. Such standards mandate strict adherence to process and specified documentation to achieve safety or security.

The SE Vee model (named due to its shape representing the letter “V” in the English language), introduced in Forsberg and Mooz (1991), described in Forsberg, et al. (2005), and shown in [Figure 2.6](#), is another example of a sequential approach used to visualize key areas for SE focus, associating each development stage with a corresponding testing stage. The Vee highlights the need for continuous validation with the stakeholders, the need to define verification plans during requirements development, and the importance of continuous risk and opportunity assessment.

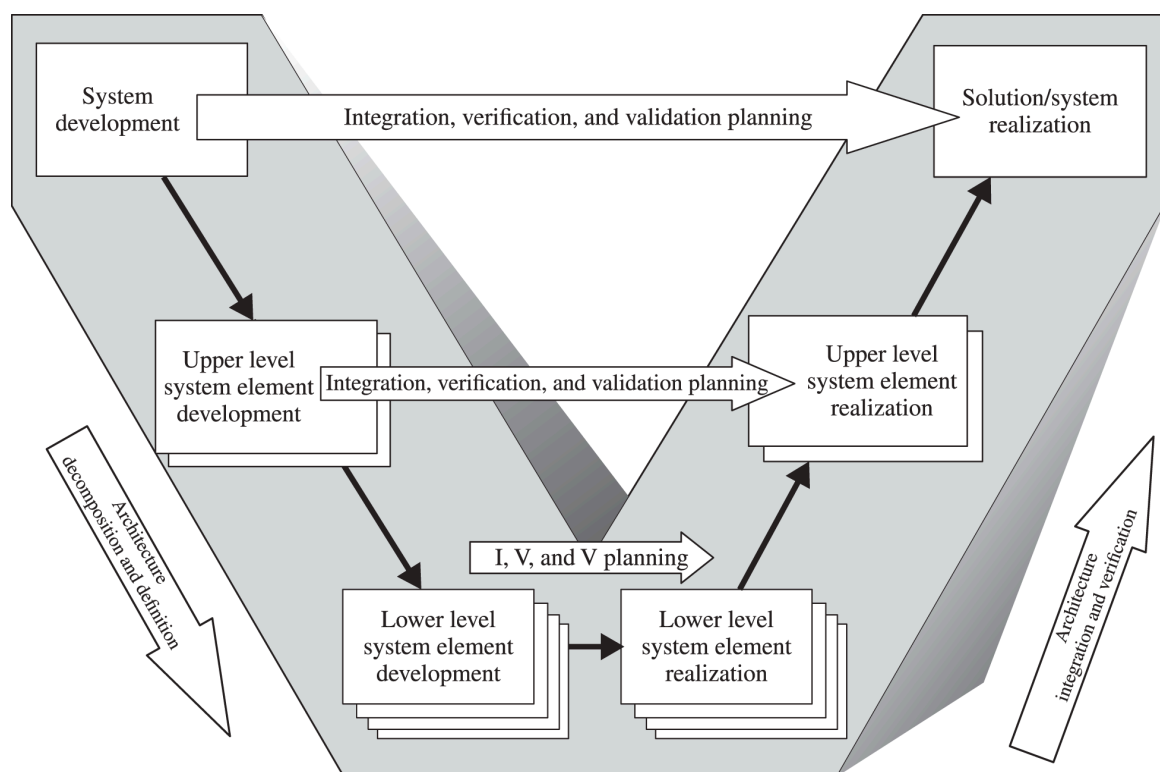


FIGURE 2.6 The SE Vee model. From Forsberg, et al. (2005) with permission from John Wiley & Sons. All other rights reserved.

There are several variations of the Vee model. Typically, the “left” side of the Vee is called system definition and the “bottom” and “right” side of the Vee are called system realization. In the Vee model, time and system maturity conceptually proceed from left to right (down the left side of the Vee and up the right side of the Vee). However, all the system life cycle processes are performed concurrently and iteratively at each level of the

system hierarchy and all the system life cycle processes are recursively applied at each level of the system hierarchy (see [Section 2.3.1.2](#)). One of the strengths of the Vee model is its depiction of the relationships between the left and right sides of the Vee.

The left side of the Vee depicts the evolving baseline from stakeholder requirements, to system requirements, to the identification of a system architecture, to definition of elements that will comprise the final system. The development team then can move from the highest level of the system requirements down to the lowest level of detail. Risk and opportunity management investigations address development options to provide assurance that the baseline performance being considered can indeed be achieved and to initiate alternate concept studies at the lower levels of detail to determine the best approach. Stakeholder discussions (in-process validation) occur to ensure that the proposed baselines are acceptable to the organization, customer, user, and other stakeholders. Changes to enhance system performance or to reduce risk or cost are welcome for consideration, but after baselining these must go through formal change control, since others may be building on previously defined and released design decisions. The bottom of the Vee depicts either the recursive application of the systems life cycle processes at the next level of the system hierarchy or the implementation of atomic system elements (see [Section 1.3.5](#)). The broadening at the base of the figure shows the growth in the number of system elements. Note that system elements can also be bought or reused. The right side of the Vee depicts the evolving baseline of system elements that are implemented, integrated, verified, and validated. In each stage of the system life cycle, the SE processes iterate to ensure that a concept or design is feasible and that the stakeholders remain supportive of the solution as it evolves.

ISO/IEC/IEEE 24748–2 (2018), Clause 6.4.3.1 provides further details on sequential life cycle models, including typical applicable systems as well as risks and opportunities associated with these models.

2.2.2 Incremental Methods

Incremental approaches have been in use since the 1960s (Larman and Basili, 2003). They represent a practical and useful approach that allows a project to provide an initial capability (or a limited set of capabilities) followed by successive deliveries to reach the desired SoI. The goal of an incremental approach is to provide rapid value and responsiveness. Generally, each increment adds capabilities intended to converge on a stakeholder satisfying result for the increment. Based on a set of requirements, a candidate set of increments is defined and the initial increment is initiated. Subsequent increments are initiated, and the process is repeated, until a complete system is deployed or until the organization decides to terminate the effort. Intermediate increments can potentially be deployed to support learning.

An incremental approach works well when an organization intends to market new versions of a product at planned intervals. Typically, the capabilities of the final delivery are known at the beginning. However, as there is significant technical risk, the development of the capabilities is performed incrementally to allow for the latest technology insertion or potential changes in needs or requirements. A core part of the planning process for an incremental approach establishes the cycle times for increments. Increments are beneficially timed in development projects to accommodate coordinated events such as integration testing and evaluation, capability deployment, experimental deployment, or release to production. Iteration cycles are beneficially timed to minimize rework cost as a project learns experimentally and empirically. Project planning and management often benefit from a constant cadence among increments.

One example of an incremental approach is the Incremental Commitment Spiral Model (ICSM) (Boehm, et al., 2014), which extends the classic Spiral Model for software introduced in Boehm (1987) for SE. A view of the ICSM is shown in [Figure 2.7](#). In the ICSM, each increment addresses requirements and solutions concurrently, rather than sequentially. ICSM also considers products and processes; hardware, software, and human aspects; and

business case analyses of alternative product configurations or product line investments. The stakeholders consider the risks and risk mitigation plans and decide on a course of action. If the risks are acceptable and covered by risk mitigation plans, the project proceeds into the next spiral (increment).

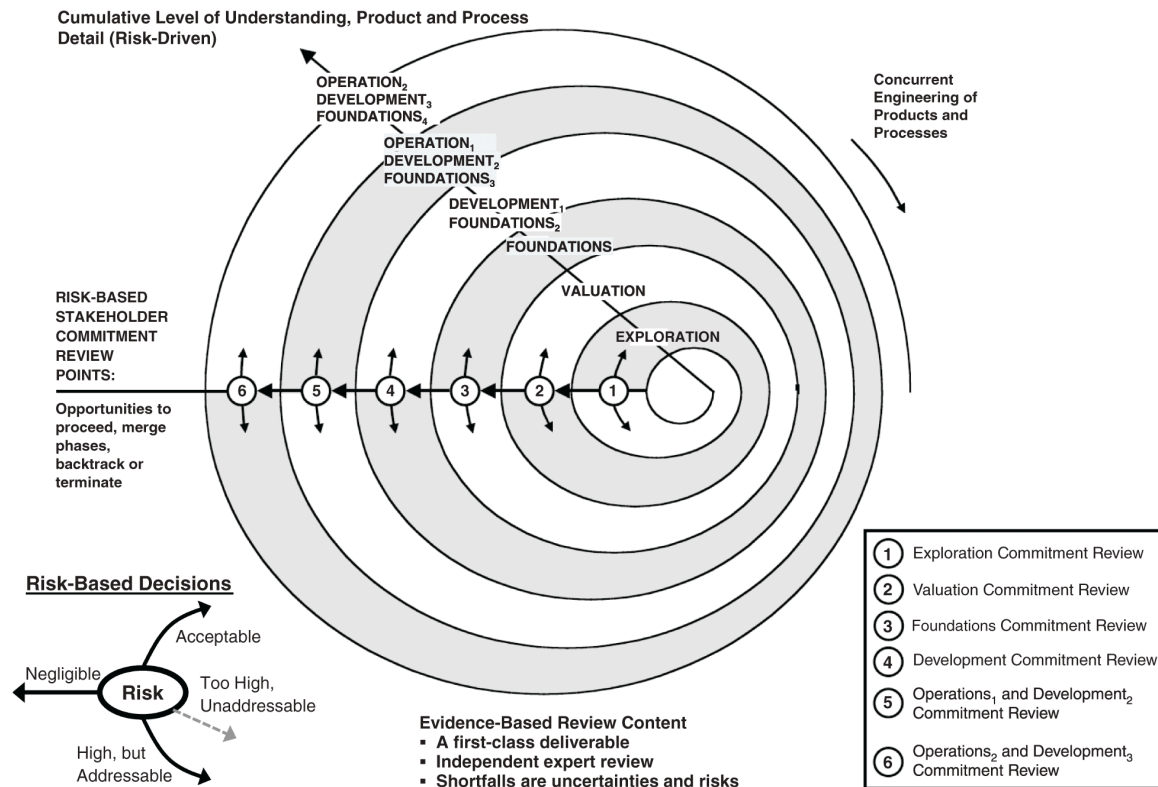


FIGURE 2.7 The Incremental Commitment Spiral model (ICSm). From Boehm, et al. (2014) with permission from pearson Education. All other rights reserved.

ISO/IEC/IEEE 24748–2 (2018), Clause 6.4.3.2 provides further details on incremental life cycle models, including typical applicable systems as well as risks and opportunities associated with these models.

2.2.3 Evolutionary Methods

In the sequential and incremental approaches described previously, the full set of required capabilities of the final system is assumed to be mostly known at the start of the effort. In some cases, especially in novel systems, the final system requirements may be unknown or only partially known. An evolutionary approach provides the adaptability and flexibility needed for the development in these situations. For example, the high-temperature tiles of the NASA space shuttle were developed using an evolutionary

approach (Forsberg, 1995). An evolutionary approach is often used in research and development (R&D) projects and SoS developments. Software development efforts are increasingly using agile methods, which are a type of evolutionary development.

In evolutionary approaches, cycles are typically planned on a regular periodic basis, each resulting in a deployable version. The requirements for the SoI are typically only partially known and are increasingly refined with each cycle. At the beginning, the goal of each cycle may be more or less unknown. Therefore, it is particularly important that the experience gained with the earlier cycles is taken into account for the subsequent cycles. Similar to the incremental approach, versions may be developed sequentially or in parallel. This is a particular challenge for those involved in the project, since new capabilities are typically assigned to exactly one version. If this assignment is lost or becomes unclear, this leads to confusion and negatively impacts the schedule and cost targets. Thus, a well-functioning configuration control is essential, also since multiple versions can be operated and supported simultaneously (see [Section 2.3.4.5](#)). Aspects to be considered include operating manuals, maintenance instructions, spare parts, disposal instructions, etc.

The evolutionary approach offers significant advantages if it is possible to obtain steady and high-quality feedback from relevant stakeholders. For example, the first versions can be used to demonstrate basic feasibility, such as a minimal viable product (MVP), and facilitate market entry. Likewise, emerging technical innovations can be planned for later versions.

When developing subsequent versions, it is recommended to carefully examine whether the previous versions should be completely replaced by newer ones. Alternatively, subsequent versions can be developed such that a partial or even complete upgrade of the previous versions to the new version is possible. For this, it is necessary that these things are considered during the early cycles. Criteria such as adaptability, flexibility, and modularity should be carefully considered to enable the long-term

evolution of the system. Decisions should be made in the context of life cycle cost (see [Section 3.1.2](#)).

An example of an evolutionary approach is DevOps (a blend of the terms and concepts for “development” and “operations”). The goal of DevOps is to provide continuous integration of the system and continuous delivery of capabilities. DevOps is typically characterized by three key principles: shared ownership, workflow automation, and rapid feedback. DevSecOps (a blend of “development,” “security,” and “operations”), shown in [Figure 2.8](#), integrates security practices into DevOps. In DevSecOps, each delivery team is responsible and empowered to pick appropriate security means.

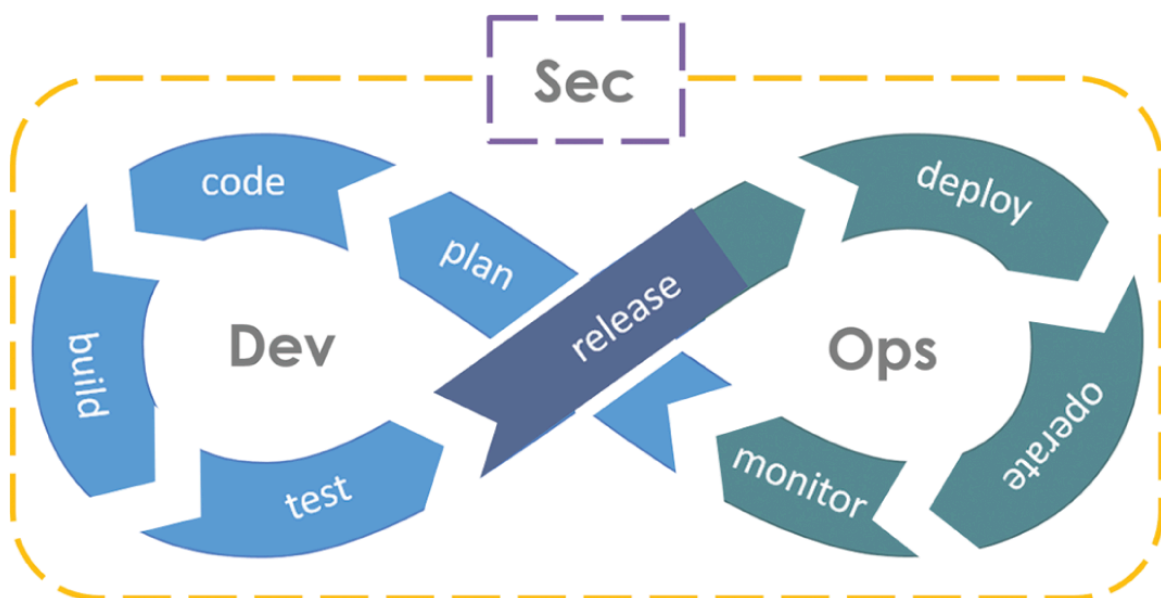


FIGURE 2.8 DevSecOps INCOSE SEH original figure created by D'Souza derived from Banach (2019) and Anx (2021). Usage per the INCOSE Notices page. All other rights reserved.

ISO/IEC/IEEE 24748–2 (2018), Clause 6.4.3.4 provides further details on evolutionary life cycle models, including typical applicable systems as well as risks and opportunities associated with these models.

[Figure 2.9](#) is an example of a mixed approach (both incremental and evolutionary). This figure shows the agile mixed-discipline approach employed by Rockwell Collins in the development of military radios (Dove, et al., 2017). Teams working on electronic-board hardware, firmware, and software have different timings for hardware increments and firmware and software epics (versions). The teams accomplish integrated work-in-

process testing with the latest increments and versions from each of the disciplines.

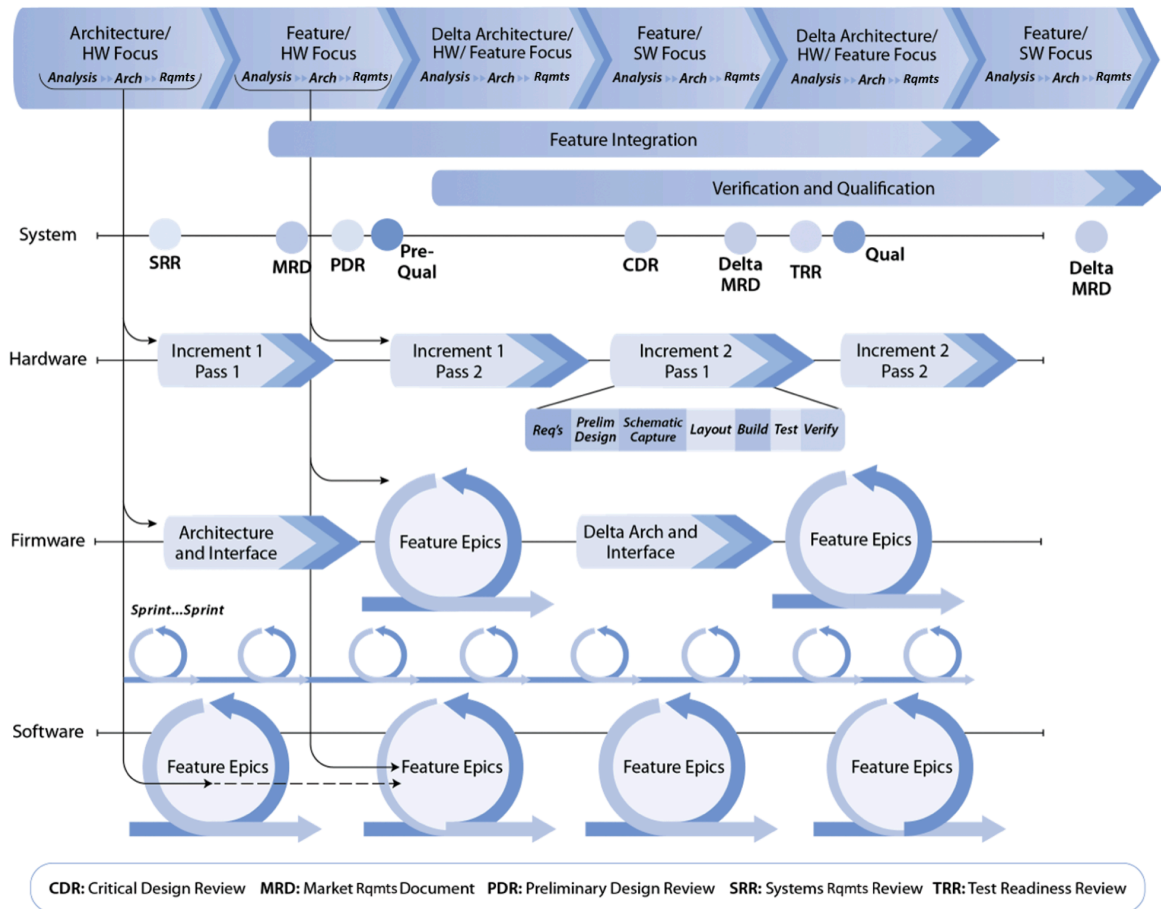


FIGURE 2.9 Asynchronous iterations and increments across agile mixed discipline engineering. From Dove, et al. (2017). Used with permission. All other rights reserved.

All these life cycle approaches are supported by the processes defined in ISO/IEC/IEEE 15288 (2023) and this handbook. The life cycle model should be chosen so that it is sufficiently adaptable and flexible. [Section 4.1](#) provides more information on tailoring life cycle models.

2.3 SYSTEM LIFE CYCLE PROCESSES

2.3.1 Introduction to the System Life Cycle Processes

A process is a series of activities and tasks performed to achieve one or more outcomes for a stated purpose. In SE, the system life cycle processes are one of the enablers to help manage a system solution across the life cycle stages. The processes are intended to be applied concurrently,

iteratively, and recursively with other enablers (e.g., tools, technology) throughout the stages of the life cycle (see [Section 2.3.1.2](#)).

ISO/IEC/IEEE 15288 (2023) identifies four process groups for the system life cycle, providing “a common process framework for describing the life cycle of engineered systems, adopting a Systems Engineering approach.” Each of these process groups is the subject of a section within Part 2. A graphical overview of these processes is given in [Figure 2.10](#):

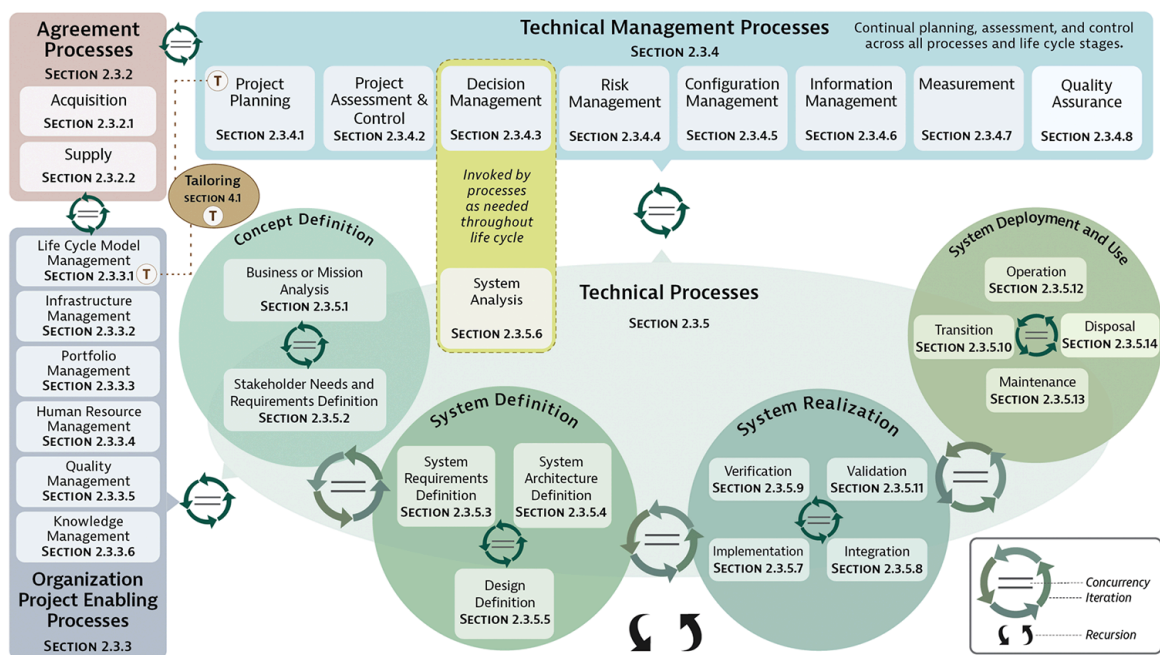


FIGURE 2.10 System life cycle processes per ISO/IEC/IEEE 15288. InCOSE SEH original figure created by roedler and Walden. Usage per the InCOSE notices page. All other rights reserved.

Agreement Processes ([Section 2.3.2](#)) include Acquisition and Supply.

Organizational Project-Enabling Processes ([Section 2.3.3](#)) include Life Cycle Model Management, Infrastructure Management, Portfolio Management, Human Resource Management, Quality Management, and Knowledge Management.

Technical Management Processes ([Section 2.3.4](#)) include Project Planning, Project Assessment and Control, Decision Management, Risk Management, Configuration Management, Information Management, Measurement, and Quality Assurance.

Technical Processes ([Section 2.3.5](#)) include Business or Mission Analysis, Stakeholder Needs and Requirements Definition, System Requirements Definition, System Architecture Definition, Design Definition, System Analysis, Implementation, Integration, Verification, Transition, Validation, Operation, Maintenance, and Disposal.

The application of these system life cycle processes is supported by SE practitioners having the relevant competencies. The competencies are defined in the INCOSE Systems Engineering Competency Framework (SECF) (2018). Note that the professional competencies (see [Section 5.1.2](#)) generally apply to all the processes.

Note: Acronyms for the process names are provided in Appendix D.

2.3.1.1 Format and Conventions

A common section structure has been applied to describe the system life cycle processes in this handbook. The following structure provides consistency in the discussion of these processes:

- Process overview
 - Purpose
 - Description
 - Inputs/outputs
 - Process activities
 - Common approaches and tips
- Process elaboration

To ensure consistency with ISO/IEC/IEEE 15288, the purpose statements from the standard are included verbatim for each process described herein. The titles of the process activities listed in each section are also consistent with ISO/IEC/IEEE 15288. The process activities describe “what” should be done, not “how” to do it. In some cases, additional items have been included to provide summary-level information regarding industry good practices and evolutions in the application of SE processes. Process elaborations provide additional details on topics that are unique to the

specific life cycle process. See Part III for topics that cross-cut multiple life cycle processes.

In addition, each system life cycle process is illustrated by an input–process–output (IPO) diagram showing typical inputs, process activities, and typical outputs. A sample is shown in [Figure 2.11](#). To understand a given process, readers are encouraged to study the complete information provided in the combination of figures and text and not rely solely on the IPO diagrams.

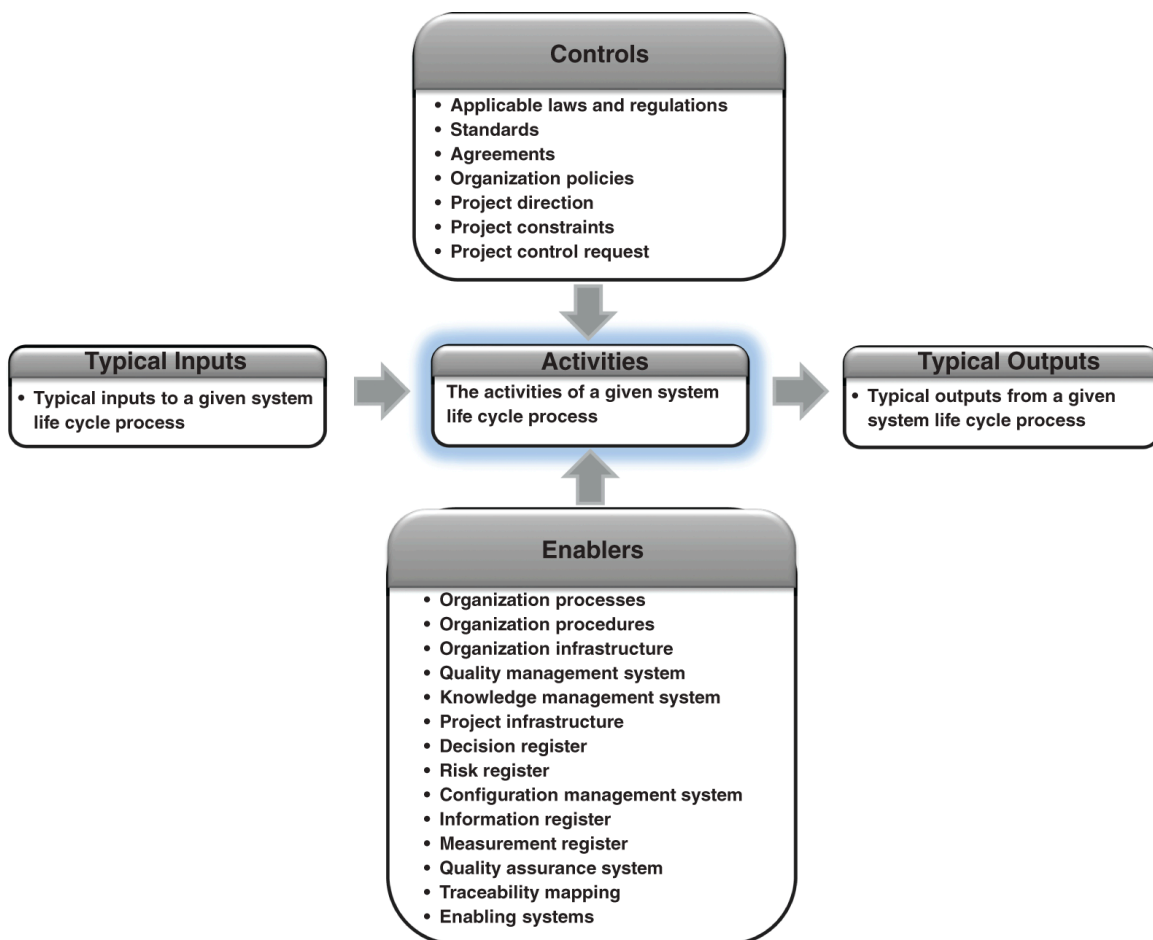


FIGURE 2.11 Sample IPO diagram for SE processes. INCOSE SEH original figure created by Walden, Shortell, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Typical inputs and outputs are listed by name within the respective IPO diagrams with which they are associated. The typical inputs and outputs are consistent with ISO/IEC 33060 (2020) when possible. Note that the IPO diagrams throughout this handbook represent “a” way that the SE processes can be performed, but not necessarily “the” way that they must

be performed. The system life cycle processes produce “results” that are often captured in “documents” or “artifacts” or “models,” rather than producing “documents” simply because they are identified as outputs. A complete list of all inputs and outputs with their respective descriptions appears in Appendix E.

The controls and enablers shown in [Figure 2.11](#) govern all processes described herein and, as such, are not repeated on the subsequent IPO diagrams. Typically, IPO diagrams do not include controls and enablers, but since they are not repeated in the IPO diagrams throughout the rest of the handbook, we have chosen to label them IPO diagrams. The enablers work together with the inputs to be transformed by the process into the outputs under the direction of the controls. A complete list of all controls and enablers with their respective descriptions appears in Appendix E.

2.3.1.2 Concurrency, Iteration, and Recursion

Too often, the system life cycle processes are viewed as being applied in a sequential, linear manner at a single level of the system hierarchy. However, valuable information and insight need to be exchanged between the processes in order to ensure a good system definition that effectively and efficiently meets the stakeholder needs and requirements. The application of concurrency, iteration, and recursion to the system life cycle processes helps to ensure communication that accounts for ongoing learning and decisions. This facilitates the incorporation of learning from further analysis and process application as the technical solution evolves. [Figure 2.12](#) shows an illustration of the concurrent, iterative, and recursive nature of the system life cycle processes.

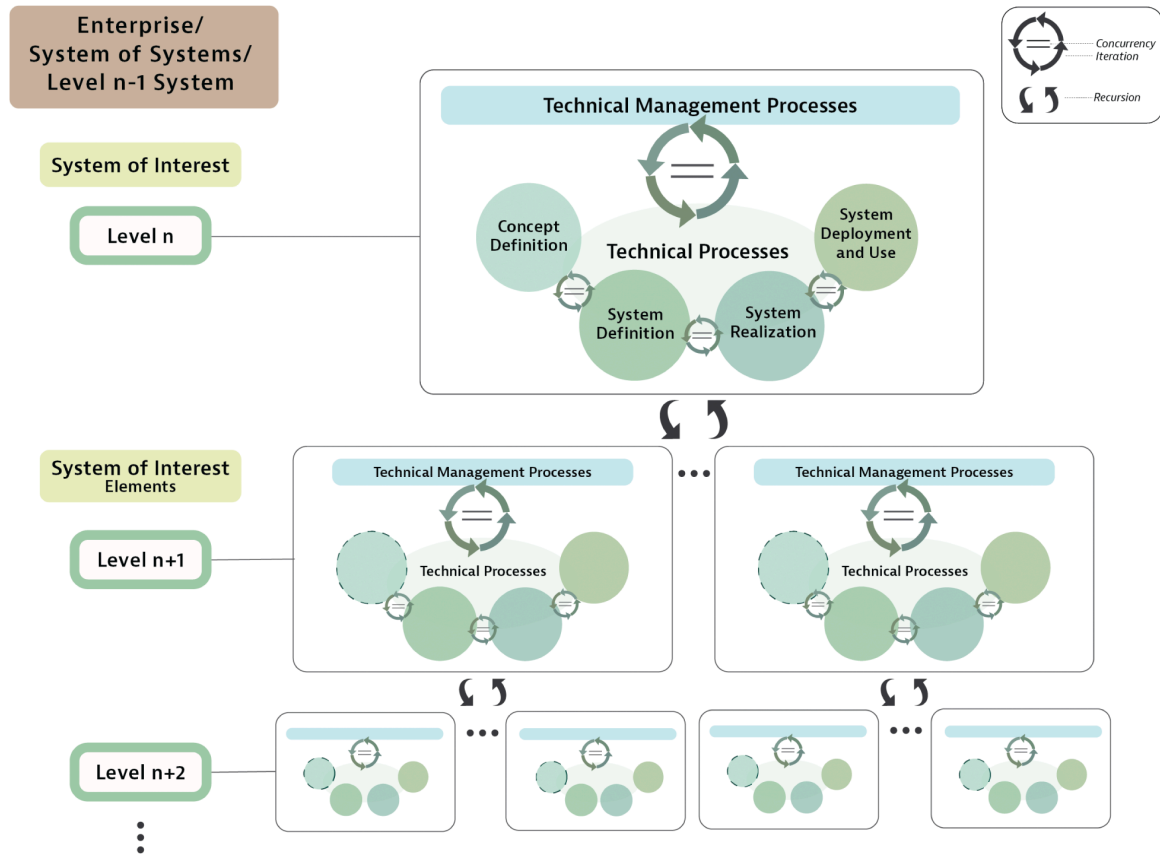


FIGURE 2.12 Concurrency, iteration, and recursion. InCOSE SEH original figure created by roedler and Walden. Usage per the InCOSE notices page. All other rights reserved.

Concurrency (indicated by the parallel lines in the figure) is the parallel application of two or more processes at a given level in the system hierarchy. Concurrent work is likely to happen on any project and the system life cycle processes can likewise be performed in a concurrent manner. It is not necessary for processes to be performed serially, especially when one process is not dependent on another for information or results. For example, the Risk Management process and Measurement process usually are performed in a continual and concurrent manner. This is illustrated in [Figure 2.34](#), in which both of these processes occur concurrently, yet provide information to one another. Additionally, the system architecture should enable concurrency through modularization, encapsulation, commonality/reuse, and other design methods.

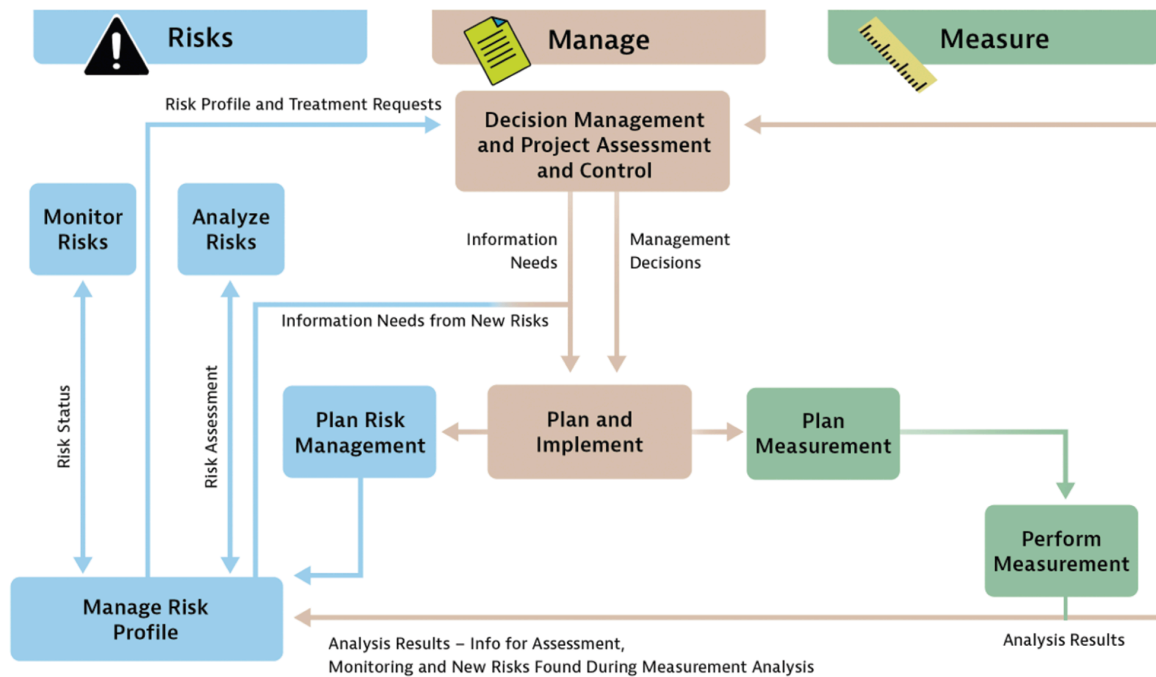


FIGURE 2.34 Integration of Measurement, Risk Management, and Decision Management processes. INCOSE SEH original figure created by Roedler. Usage per the INCOSE Notices page. All other rights reserved.

Iteration (indicated by the circular arrows in the figure) is the repeated application of and interaction between two or more processes at a given level in the system hierarchy. Iteration is needed to accommodate stakeholder decisions and evolving understanding, account for architectural decisions or constraints, and resolve trade-offs for affordability, adaptability, feasibility, resilience, etc. There can be iteration between any of the processes. For example, there is often iteration between System Requirements Definition and System Architecture Definition processes. The system architecture will reflect these design iterations through identification of functions, their allocation to system elements, assignment to logical and physical interfaces, and verification as intended in the design. In this case, there is a concurrent application of the processes with iteration between them, where the evolving system requirements help to shape the architecture through identified constraints and functional and quality requirements. For example, the system architecture may need to be changed due to detailed electrical modeling indicating that a particular system element's load exceeds its allocated power budget and forces a design change or reallocation of the network power assignments in the overall system. The tradeoffs between candidate architectures or elements of the architecture, in turn, may identify requirements that are not feasible,

driving further requirements analysis with trade-offs that change some requirements. Likewise, the Design Definition process could identify the need to reconsider decisions and trade-offs in the System Requirements Definition or System Architecture Definition processes. Any of these can invoke additional application of the System Analysis and Decision Management processes.

Recursion (indicated by the down and up arrows in the figure) is the repeated application of the set of life cycle processes, tailored as appropriate, at successive levels in the system hierarchy. The Technical Management and Technical Processes are expected to be recursively applied for each successive level of the system hierarchy until the level is reached where the decision is made to make, buy, or reuse a system element (see [Section 1.3.5](#)). During the recursive application of the processes, the outputs at one level become inputs for the next successive level (below for system definition, above for system realization).

Horizontal integration ensures completeness before diving deeper and *vertical integration* ensures consistency between levels in this concurrent, iterative, and recursive environment (see [Section 2.3.5.8](#)). For example, one may have to define functions, their inputs and outputs, their associated performance and conditions of operations *before* writing the associated requirement. Then one can define the related verification (method, conditions, criteria), from which one can postulate the next-lower-level architecture to assess feasibility and perform the related function, performance, or requirements decomposition and allocation. However, teams may need to go down multiples levels to validate that the functions or elements at the lower levels are going to be suitable solutions for the SoI and its stakeholders.

2.3.2 Agreement Processes

The initiation of a project begins with the identification of a problem or opportunity to be addressed, which results in the development of needs to be satisfied. Once a need is identified and resources are committed to

establish a project, it is possible to define the terms and conditions of an acquisition and supply relationship through the Agreement Processes, which are defined in ISO/IEC/IEEE 15288 as follows:

[5.7.2] [Agreement] Processes define the activities necessary to establish an agreement between two organizations.

The Agreement Processes in this handbook and in ISO/IEC/IEEE 15288 are focused on the acquisition and supply of systems, system elements, products, or services, although agreements could be established for other objectives. With respect to the acquisition and supply relationship, the acquirer and supplier could be two independent organizations (i.e., no common parent organization or enterprise) or two organizations from the same parent organization or enterprise.

The Agreement Processes are utilized under many conditions, including when:

- an organization cannot satisfy a defined need itself,
- a supplier can satisfy a defined need in a more economical or timely manner,
- a higher authority has directed the use of a specific supplier, and
- an organization needs materials or specialized services.

An overall objective of Agreement Processes is to identify the interfaces between the acquirer and supplier(s) and establish the terms and conditions of these relationships, including identifying the inputs required and the outputs that will be provided.

Agreement negotiations are handled in various ways depending on the specific organizations and the formality of the agreement. In a formal agreement, there is usually a contract negotiation activity to refine the contract terms and conditions. Note also that the Agreement Processes can be used for coordinating within an organization between different business

units or functions. In this case, the agreement will usually be more informal, not requiring a formal or specific contract.

An important contribution of ISO/IEC/IEEE 15288 is the recognition that SE practitioners are relevant contributors to the Agreement Processes (Arnold and Lawson, 2003). The SE practitioner is usually in a supporting role to the project management practitioner during negotiations and is responsible for impact assessments for changes, trade studies on alternatives, risk assessments, and other technical input needed for decisions.

Acceptance criteria are critical elements to each party because they protect both sides of the business relationship—the acquirer from being coerced into accepting a product with poor quality and the supplier from the unpredictable actions of an indecisive acquirer. It is important to note that the acceptance criteria are negotiated during the Agreement Processes. During negotiations, it is also critical that both parties are able to track progress toward an agreement. Identifying where further work toward achieving consensus in the documents and clauses is vital.

Two Agreement Processes are identified by ISO/IEC/IEEE 15288: the Acquisition process and the Supply process. These processes, subject of [Sections 2.3.2.1](#) and [2.3.2.2](#), respectively, are two sides of the same coin. They conduct the essential business of the organization related to the SoI. They establish the relationships between organizations relevant to the acquisition and supply of products and services, regardless of whether the agreement is formal (as in a contract) or informal. Each process establishes the context and constraints of the agreement under which the other system life cycle processes belonging to the project scope are performed. Note that an organization can be both a supplier and acquirer for a given system. For example, an organization may be contracted to supply a system to an end customer. However, that organization may choose to acquire some of the system elements, materials, or services for developing or producing the system. So, that organization is the supplier to the end customer of the system and is the acquirer to those organizations providing it system elements, materials, or services.

Changes may happen during the execution of an agreement including acquirer change requests, deviations and waivers from the supply chain, or changes in the context of the project that were foreseen in risk analysis or not. Upon decision of the parties, this may lead to modifications to the initial state of the agreement. For that purpose, a statement of compliance may be initiated and updated all along the project describing the agreed changes and can include requirements impacted by a modification, the reference of modification, compliance verification by the supplier, and compliance validation by the acquirer.

2.3.2.1 Acquisition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.1.1.1] The purpose of the Acquisition process is to obtain a product or service in accordance with the acquirer's requirements.

The Acquisition process is invoked to establish an agreement between two organizations under which one party acquires products and/or services from the other. The acquirer experiences a need for an operational system, for services in support of an operational system, for elements of a system being developed by a project, or for services in support of project activities.

Description This section is written from the perspective of the acquirer organization. An acquiring organization applies due diligence in the selection of a supplier to avoid costly failures and impacts to the organization's budgets and schedules and other issues. Therefore, the role of the acquirer demands familiarity with the Technical, Technical Management, and Organizational Project-Enabling Processes, as it is through them that the supplier will execute the agreement.

Inputs/Outputs Inputs and outputs for the Acquisition process are listed in [Figure 2.13](#). Descriptions of each input and output are provided in

Appendix E.

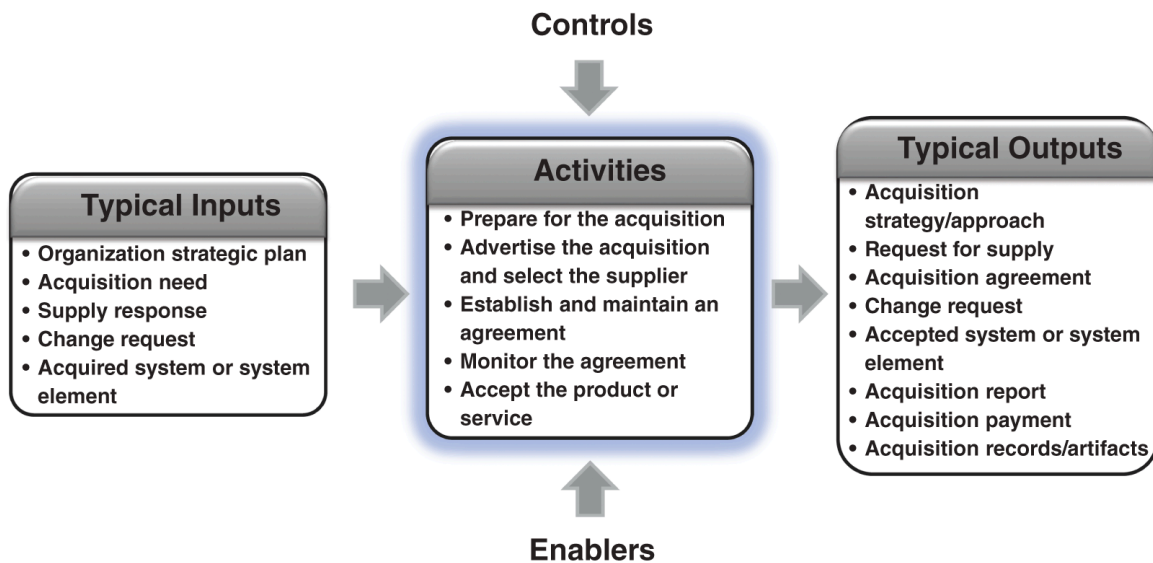


FIGURE 2.13 IPO diagram for the Acquisition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Acquisition process includes the following activities:

- *Prepare for the acquisition.*
 - – Develop and maintain acquisition policies, plans, and procedures to meet the organization strategies, goals, and objectives as well as the needs of the project management and SE organizations.
 - – Collect needs in a request for supply—such as a Request for Proposal (RFP) or Request for Quotation (RFQ) or some other mechanism—to obtain the supply of the service and/or product. Through the use of the Technical Processes, the acquiring organization produces a set of requirements and models that will form the basis for the technical information of the agreement.
 - – Identify a list of potential suppliers—suppliers may be internal or external to the acquirer organization.
- *Advertise the acquisition and select the supplier.*
 - – Distribute the request for supply and select appropriate suppliers—using selection criteria, rank suppliers by their suitability to meet the overall need and establish supplier preferences and corresponding justifications. Viable suppliers should be willing to conduct ethical negotiations, able to meet obligations, and willing to maintain open

communications throughout the Acquisition process. Note that the approach may be less formal when a function within the organization is a candidate for the supply need.

- – Evaluate supplier responses to the request for supply—ensure the offered product and/or service can meet acquirer needs and complies with industry and other standards. Assessments from the Project Portfolio Management (see [Section 2.3.3.3](#)) and Quality Management (see [Section 2.3.3.5](#)) processes and review results from the requesting organization are necessary to determine the suitability of each response and the ability of the supplier to meet the stated commitments. Record results from the evaluation of responses to the request for supply. This can range from formal documentation to less formal interorganizational interactions (e.g., between design engineering and marketing).
- – Select the preferred supplier(s) based on acquisition criteria.
- *Establish and maintain an agreement.*
 - – Establish an agreement. Ensure an understanding of expectations, including acceptance criteria.
 - – This agreement ranges in formality from a written contract to a verbal agreement. Appropriate to the level of formality, the agreement establishes requirements, development and delivery milestones, verification, validation and acceptance conditions, process requirements (e.g., configuration management, risk management, measurements), exception-handling procedures, agreement change management procedures, payment schedules, and handling of data rights and intellectual property so that both parties understand the basis for executing the agreement. For a written contract, this occurs when the contract is signed.
 - – Identify the necessary changes to the agreement and evaluate the related impacts on the agreement.
 - – Update the agreement with the supplier as necessary.
- *Monitor the agreement.*
 - – Manage Acquisition process activities, including decision making for agreements, relationship building and maintenance, interaction with organization management, responsibility for the development

of plans and schedules, and final approval authority for deliveries accepted from the supplier.

- – Maintain communications with supplier, stakeholders, and other organizations regarding the project.
 - – Status progress against the agreed-to schedule to identify risks and issues, to measure progress toward mitigation of risks and adequacy of progress toward delivery and cost and schedule performance, and to determine potential undesirable outcomes for the organization.
- The Project Assessment and Control process (see [Section 2.3.4.2](#)) provides necessary evaluation information regarding cost, schedule, and performance.
- – Amend agreements when impacts on schedule, budget, or performance are identified.
 - *Accept the product or service.*
 - – Accept delivery of products and services—in accordance with all agreements and relevant laws and regulations.
 - – Render payment—or other agreed consideration in accordance with agreed payment schedules.
 - – Accept responsibility in accordance with all agreements and relevant laws and regulations.
 - – When an Acquisition process cycle concludes, a final review of performance is conducted to extract lessons learned for continued process performance.
 - – Retire the agreement.

Note: The project is closed through the Portfolio Management process (see [Section 2.3.3.3](#)), which manages the full set of projects of the organization.

Common approaches and tips:

- Establish acquisition guidance and procedures that inform acquisition planning, including recommended milestones, standards, assessment criteria, and decision gates. Include approaches for identifying, evaluating, choosing, negotiating, managing, and terminating suppliers.

- Establish a technical point of responsibility within the organization for monitoring and controlling individual agreements. This person maintains communication with the supplier and is part of the decision-making team to assess technical development and progress in the execution of the agreement.

Note: There can be multiple points of responsibility for an agreement that focus on technical, programmatic, marketing, etc.

- Define and track measures that indicate progress on agreements. Avoid measures that are not focused on the true information needs. Leading indicators are preferable (see [Section 2.3.4.7](#)).
- Include technical representation in the selection of the suppliers to critically assess the capability of the supplier to perform the required task.
- Past performance of the supplier is highly important, but changes to key supplier personnel should be identified and evaluated to understand any impact with respect to the current request for supply.
- Communicate clearly with the supplier about priorities and avoid conflicting statements or making frequent changes in the statement of need that introduce risk into the process.
- Maintain traceability between the supplier's responses to the acquirer's solicitation. This can reduce the risk of contract modifications, cancellations, or follow-on contracts to fix the product or service.
- Smart contracts can be used to establish and maintain an agreement. A smart contract is a transaction protocol intended to execute automatically and control or document legally relevant events and actions according to the terms of a contract or an agreement (Tapscott and Tapscott, 2018). The objectives of smart contracts are the reduction of need in trusted intermediaries, arbitrations and enforcement costs, fraud losses, and the reduction of malicious and accidental exceptions (Fries and Paal, 2019).

Elaboration:

The Project Manager's role is to define, execute, and manage the acquisition. This is focused on the project needs to deliver the system, system elements, products, or services that meet the end user requirements and achieve the acquisition milestones. This is done in collaboration with the SE practitioners and the selected contractor to ensure the technical expectations and key performance parameters are achieved. The team needs to define plans and methods collectively, and refine them as more is learned about the nature and challenges inherent in the system or capability being built and its intended operating environment. For more information on PM-SE integration, see 5.3.3.

When the acquisition involves systems or system elements where technology or a system capability is not mature enough, it is necessary to account for uncertainty and the need for additional risk management actions in the planning. This includes allowing additional margin in the development/production timeframe, such as ample lead time in anticipation of inherent challenges, especially when technology maturation is required. These challenges may also include limited availability of adequate resources for the supplier (skilled labor and/or technologies), a need for customization of supplier products or equipment, poor or early understanding of interface requirements, integration challenges, and required verification and/or validation of the development. If there is no flexibility in the delivery date, then trade-offs may be needed to provide the system capabilities in an incremental manner.

Technical supplier management is about ensuring the supplier meets the allocated project requirements and that the supplier is effectively managed. This is usually achieved through the Statement of Work (SOW) and a set of requirements. The SOW is a mechanism to ensure progress is being made and describes the necessary work, quality, standards, designs, models, evidence, reviews, timescales, and meetings, etc. that the supplier is expected to provide contingent on the contract. To prove the system/system element functional, performance, and operational requirements are met, the supplier will also need to provide compliance matrices and verification and validation evidence.

2.3.2.2 Supply Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.1.2.1] The purpose of the Supply process is to provide an acquirer with a product or service that meets agreed requirements.

The Supply process is invoked to establish an agreement between two organizations under which one party supplies products or services to the other. Within the supplier organization, a project is conducted according to the recommendations of this handbook with the objective of providing a product or service to the acquirer that meets the agreed requirements. In the case of a mass-produced commercial product or service, a marketing, or similar, function may represent the acquirer and establish stakeholder expectations.

Description This section is written from the perspective of the supplier organization. The Supply process is highly dependent upon the Technical, Technical Management, and Organizational Project-Enabling Processes as it is through them that the work of executing the agreement is accomplished. This means that the Supply process is the larger context in which the other processes are applied under the agreement.

Inputs/Outputs Inputs and outputs for the Supply process are listed in [Figure 2.14](#). Descriptions of each input and output are provided in Appendix E.

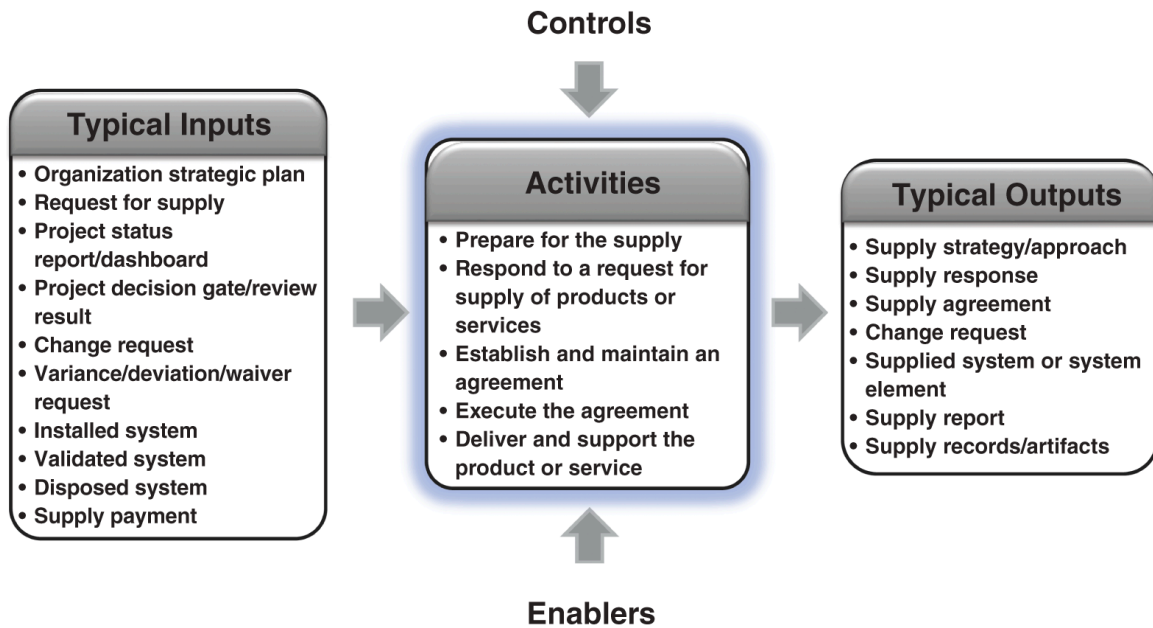


FIGURE 2.14 IPO diagram for the Supply process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Supply process includes the following activities:

- *Prepare for the supply.*
 - – Develop and maintain strategic plans, policies, and procedures to meet the needs of potential acquirer organizations, as well as internal organization goals and objectives including the needs of the project management and technical SE organizations.
 - – Identify opportunities.
- *Respond to a request for supply of products or services.*
 - – Select appropriate acquirers willing to conduct ethical negotiations, able to meet financial obligations, and willing to maintain open communications throughout the Supply process.
 - – Evaluate the acquirer requests and propose a product or service that meets acquirer needs and complies with industry and other standards. Assessments from the Portfolio Management, Human Resource Management, Quality Management, and Business or Mission Analysis processes are necessary to determine the suitability of this response and the ability of the organization to meet these commitments.
- *Establish and maintain an agreement.*

- – Establish an agreement. Ensure an understanding of expectations, including acceptance criteria.
- – Identify the necessary changes to the agreement and evaluate the related impacts on the agreement.
- – Update the agreement with the acquirer as necessary.
- *Execute the agreement.*
 - – Start the project and invoke the other processes defined in this handbook.
 - – Manage the Supply process and related activities including the development of plans and schedules, decision making for agreements, relationship building and maintenance, interaction with organization management, and final approval authority for deliveries made to acquirer.
 - – Maintain communications with acquirers, suppliers, stakeholders, and other organizations regarding the agreement.
 - – Carefully evaluate the terms of the agreement to identify risks and issues, progress toward mitigation of risks, and adequacy of progress toward delivery. Also evaluate cost and schedule performance and determine potential undesirable outcomes for the organization.
- *Deliver and support the product or service.*
 - – After acceptance and transfer of the final products and/or services, the acquirer will provide payment or other consideration in accordance with all agreements, schedules, and relevant laws and regulations. A support agreement is often ongoing after the transfer of products and/or services.
 - – When a Supply process cycle concludes, a final review of performance is conducted to extract lessons learned for continued process performance.
 - – Retire the agreement.

Note: The agreement is closed through the Portfolio Management process (see Section [2.3.3.3](#)), which manages the full set of systems and projects of the organization. When the project is closed, action is taken to close the agreement.

Common approaches and tips:

- Relationship building and trust between the parties is a nonquantifiable quality that, while not a substitute for good processes, makes human interactions agreeable.
- Develop technology white papers or similar artifacts to demonstrate and describe to the (potential) acquirer the range of capabilities in areas of interest. Use traditional marketing approaches to encourage acquisition of mass-produced products.
- When expertise is not available within the organization (legal and other governmental regulations, laws, etc.), retain subject matter experts to provide information and specify requirements related to agreements.
- Invest sufficient time and effort into understanding acquirer needs before the agreement. This can improve the estimations for cost and schedule and positively affect agreement execution. Evaluate any technical specifications for the product or service for clarity, completeness, and consistency.
- Involve personnel who will be responsible for agreement execution to participate in the evaluation of and response to the acquirer's request. This reduces the start-up time once the project is initiated, which in turn is one way to recapture the cost of writing the response.
- Make a critical assessment of the ability of the organization to execute the agreement; otherwise, the high risk of failure and its associated costs, delivery delays, and increased resource commitment needs will reflect negatively on the reputation of the entire organization.

Elaboration:

Agreements fall into a large range, from formal to very informal based on verbal understanding (e.g., from a written contract to a verbal agreement). Agreements may call for a fixed price, cost plus fixed fee, incentives for early delivery, penalties for late deliveries, and other financial motivators. Appropriate to the level of formality, the agreement establishes requirements, development and delivery milestones, verification, validation and acceptance conditions, process requirements (e.g.,

configuration management, risk management, measurements), exception-handling procedures, agreement change management procedures, payment schedules, and handling of data rights and intellectual property so that both parties understand the basis for executing the agreement. For a written contract, this occurs when the contract is signed.

2.3.3 Organizational Project-Enabling Processes

The Organizational Project-Enabling Processes are defined in ISO/IEC/IEEE 15288 as follows:

[5.7.3] The Organizational Project-Enabling Processes are concerned with providing the resources needed to enable the project to meet the needs and expectations of the organization's stakeholders. The Organizational Project-Enabling Processes are typically concerned at a strategic level with the management and improvement of the organization's undertaking, with the provision and deployment of resources and assets, and with its management of risks in competitive or uncertain situations.... The Organizational Project-Enabling Processes establish the environment in which projects are conducted.

This section focuses on the capabilities of an organization relevant to enabling the system life cycle; they are not intended to address general business management objectives, although sometimes the two overlap. Six Organizational Project-Enabling Processes are identified by ISO/IEC/IEEE 15288. They are Life Cycle Model Management, Infrastructure Management, Portfolio Management, Human Resource Management, Quality Management, and Knowledge Management. As defined in ISO/IEC/IEEE 15288 and this handbook, these processes provide the resources and organizational support to enable the projects that are focused on the system life cycle. The organization will tailor these processes and their interfaces to meet specific strategic and tactical objectives in support of the organization's projects (see [Section 4.1](#)).

2.3.3.1 Life Cycle Model Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.2.1.1] The purpose of the Life Cycle Model Management process is to define, maintain, and help ensure availability of policies, life cycle processes, life cycle models, and procedures for use by the organization with respect to the scope of ISO/IEC/IEEE 15288.

Description This process (i) establishes and maintains a set of policies and procedures at the organization level that support the organization's ability to acquire and supply products and services and (ii) provides integrated system life cycle models necessary to meet the organization's strategic plans, policies, goals, and objectives for all projects and all system life cycle stages. The processes are defined, adapted, and maintained to support the requirements of the organization, SE organizational units, individual projects, and personnel. The Life Cycle Model Management process is supplemented by recommended methods and tools. The resulting guidelines in the form of organization policies and procedures are still subject to tailoring by projects (see [Section 4.1](#)).

Inputs/Outputs Inputs and outputs for the Life Cycle Model Management process are listed in [Figure 2.15](#). Descriptions of each input and output are provided in Appendix E.

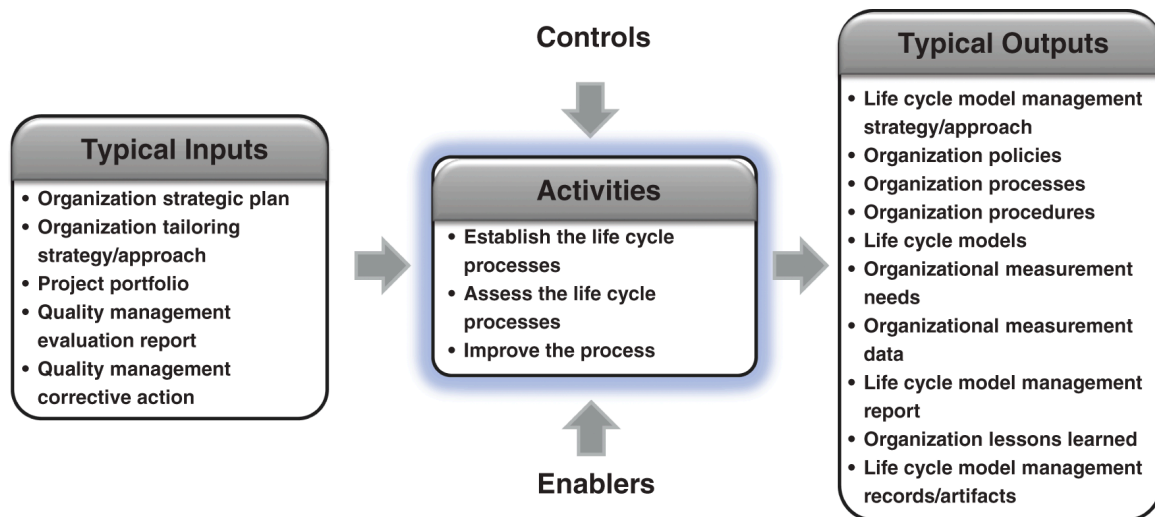


FIGURE 2.15 IPO diagram for Life Cycle Model Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Life Cycle Model Management process includes the following activities:

- *Establish the life cycle process.*
 - – Establish policies and procedures for managing and deploying life cycle processes.
 - – Establish the life cycle processes with process performance metrics to assess effectiveness and efficiency.
 - – Define roles, responsibilities, accountabilities, and authorities to enable the implementation of the life cycle processes.
 - – Establish entrance and exit criteria for decision gates.
 - – Define an appropriate set of life cycle models that are comprised of stages.
 - – Establish tailoring guidance for projects
- *Assess the life cycle process.*
 - – Use assessments and reviews of the life cycle models' performance to confirm the adequacy and effectiveness of the Life Cycle Model Management process.
 - – Identify opportunities to improve the organization life cycle model management guidelines on a continuing basis based on individual project assessments, individual feedback, metrics, and changes in the organization strategic plan.
- *Improve the process.*

- – Prioritize and implement the identified improvement opportunities.
- – Communicate with all relevant organizations regarding the creation of and changes in the life cycle model management guideline.

NOTE: ISO/IEC/IEEE 15288 provides more details for the Life Cycle Model Management process that are aligned with the activities listed above.

Common approaches and tips:

- Base the policies and procedures on an organization-level strategic and business area plan that provides a comprehensive understanding of the organization's goals, objectives, stakeholders, competitors, future business, and technology trends.
- Ensure that policy and procedure compliance review is included as part of the business decision gate criteria.
- Develop a Life Cycle Model Management process information database with essential information that provides an effective mechanism for disseminating consistent guidelines and providing announcements about organization-related topics, as well as industry trends, research findings, and other relevant information. This provides a single point of contact for continuous communication regarding the life cycle model management guidelines and encourages the collection of valuable feedback, metrics, and the identification of organization trends.
- Establish an organization center of excellence for the Life Cycle Model Management process. This organization can become the focal point for the collection of relevant information, dissemination of guidelines, and analysis of assessments, performance, and feedback. They can also develop checklists and other templates to support project assessments to ensure that the predefined measures and criteria are used for evaluation.
- Manage the network of external relationships by assigning personnel to identify standards, industry and academia research, and other sources of organization management information and concepts needed by the

organization. The network of relationships includes government, industry, and academia. Each of these external interfaces provides unique and essential information for the organization to succeed in business and meet the continued need and demand for improved and effective systems and products for its stakeholders. It is up to the Life Cycle Model Management process to fully define and utilize these external entities and interfaces (i.e., their value, importance, and capabilities that are required by the organization):

- – Legislative, regulatory, and other government requirements
- – Industry SE and management-related standards, training, and capability maturity models
- – Academic education, research results, future concepts and perspectives, and requests for financial support
- Establish an organization communication plan for the policies and procedures. Most of the processes in this handbook include dissemination activities. An effective set of communication methods is needed to ensure that all stakeholders are well informed.
- Include stakeholders, such as engineering and project management organizations, as participants in developing the life cycle model management guidelines. This increases their commitment to the recommendations and incorporates a valuable source of organizational experience.
- Develop alternative life cycle models based on the type, scope, complexity, and risk of a project. This decreases the need for tailoring by engineering and project organizations.

Elaboration

Value Proposition for Organizational Processes. The value propositions to be achieved by instituting organization-wide processes for use by projects are as follows:

- Provide repeatable/predictable performance across the projects in the organization (this helps the organization in planning and estimating future projects and in demonstrating reliability to stakeholders)

- Leverage practices that have been proven successful by certain projects and instill those in other projects across the organization (where applicable)
- Enable process improvement across the organization
- Improve ability to efficiently transfer staff across projects as roles are defined and performed consistently
- Enable leveraging lessons that are learned from one project for future projects to improve performance and avoid issues
- Improve startup of new projects (less reinventing the wheel)

In addition, the standardization across projects may enable cost savings through economies of scale for support activities (tool support, process documentation, etc.).

Benchmarking. SE benchmarking involves comparing an organization's system life cycle processes and practices to those of other entities that are considered as good performers, internally or externally, or comparing to industry standards or good practices. SE benchmarking results and comparisons can be used to generate ideas for driving process improvement to maximize efficiency and effectiveness.

Standard SE Processes. An organization engaged in SE provides the requirements for establishing, maintaining, and improving the standard SE processes and the policies, practices, and supporting functional processes necessary to meet the needs throughout the organization. Further, it defines the process for tailoring the standard SE processes for use on projects addressing the specific needs of the project and for making improvements to the project-tailored SE processes.

Analysis of Process Performance. A high-performing organization also reviews the process (as well as work products), conducts assessments and audits (e.g., assessments based on CMMI (2018), ISO/IEC 33060 (2020), and ISO 9000 (2015) audits), retains corporate memory through the understanding of lessons learned, and establishes how benchmarked processes and practices of related organizations can affect the organization.

Successful organizations should analyze their process performance, its effectiveness and compliance to organizational and higher directed standards, and the associated benefits and costs and then develop targeted improvements.

The basic requirements for standard and project-tailored SE process control, based on CMMI (2018), ISO/IEC 33060 (2020), or other resources, are as follows:

- Process responsibilities for projects:
 - – Identify SE processes.
 - – Document the implementation and maintenance of SE processes.
 - – Use a defined set of standard methods and techniques to support the SE processes.
 - – Apply accepted tailoring guidelines to the standard SE processes to meet project-specific needs.
- Good process definition includes:
 - – Inputs and outputs
 - – Entrance and exit criteria
- Process responsibilities for organizations and projects:
 - – Assess strengths and weaknesses in the SE processes.
 - – Compare the SE processes to benchmark processes used by other organizations.
 - – Institute SE process reviews and audits of the SE processes.
 - – Institute a means to capture and act on lessons learned from SE process implementation on projects.
 - – Institute a means to analyze potential changes for improvement to the SE processes.
 - – Institute measures that provide insight into the performance and effectiveness of the SE processes.
 - – Analyze the process measures and other information to determine the effectiveness of the SE processes.

Although it should be encouraged to identify and capture lessons learned throughout the performance of every project, the SE organization must

plan and follow through to collect lessons learned at predefined milestones in the system life cycle. The SE organization should periodically review lessons learned together with the measures and other information to analyze and improve SE processes and practices. The results need to be communicated and incorporated into training. It should also establish good practices and capture them in an easy-to-retrieve form.

For more information on process definition, assessment, and improvement, see the resources in the bibliography, including the CMMI and ISO/IEC TS 33060.

2.3.3.2 Infrastructure Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.2.2.1] The purpose of the Infrastructure Management process is to provide the infrastructure and services to projects to support organization and project objectives throughout the life cycle.

Description The work of the organization is accomplished through projects, which are conducted within the context of the infrastructure environment. This infrastructure needs to be defined and understood within the organization and the project to ensure alignment of the working units and achievement of overall organization strategic objectives. This process exists to establish, communicate, and continuously improve the system life cycle process environment.

Infrastructure Management is an organizational project-enabling process and foundational to all SE process management and improvement. Effective infrastructure management is imperative to an organization's ability to change and for that change to be positive, durable, and impactful. Each element of infrastructure is a SoI and both Technical Management and Technical Processes, as stated in ISO/IEC/IEEE 15288 apply to the

establishment and maintenance of the infrastructure. Additionally, the Infrastructure Management process includes the physical, political, and process improvement infrastructures.

Inputs/Outputs Inputs and outputs for the Infrastructure Management process are listed in [Figure 2.16](#). Descriptions of each input and output are provided in Appendix E.

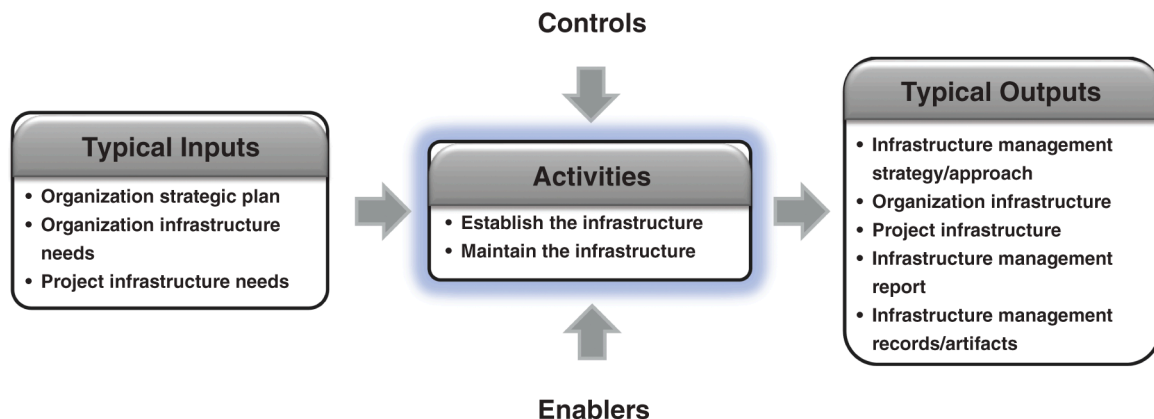


FIGURE 2.16 IPO diagram for Infrastructure Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Infrastructure Management process includes the following activities:

- *Establish the infrastructure.*
 - – Define infrastructure requirements.
 - – Define infrastructure elements (e.g., facilities, tools, hardware, software, services, and standards)
 - – Define, gather, and negotiate infrastructure resource needs with the organization and projects.
 - – Identify, obtain, and provide the infrastructure resources and services to ensure organization goals and objectives are met.
 - – Control the infrastructure elements, resources, and services.
 - – Conduct inventory management to include enumeration, lists, storage to establish ownership, accessibility, and expectations.
 - – Manage resource and service conflicts and shortfalls with steps for resolution.

- – Conduct infrastructure management inventories including identification, status, type, location, access, and condition.
- *Maintain the infrastructure.*
 - – Continue to assess whether the project infrastructure needs are met.
 - – Identify and provide improvements or changes to the infrastructure resources as the project requirements change.
 - – Manage infrastructure resource availability to ensure organization goals and objectives are met. Conflicts and resource shortfalls are managed with steps for resolution.
 - – Allocate infrastructure resources and services to support all projects.
 - – Evaluate the condition of the infrastructure.
 - – Perform cost analysis toward the cost of infrastructure management.
 - – Control multi-project infrastructure resource management communications to effectively allocate resources throughout the organization; and identify potential future or existing conflict issues and problems with recommendations for resolution.
 - – Provide change control for the infrastructure management.
 - – Conduct risk analysis regarding infrastructure management.
 - – Evaluate infrastructure management alternatives through analysis of alternatives. This evaluation and analysis compliments risk management and cost reduction activities.

Common approaches and tips:

- Qualified resources may be leased (insourced or outsourced) or licensed in accordance with the investment strategy.
- Establish an organization infrastructure architecture. Integrating the infrastructure of the organization can make the execution of routine business activities more efficient.
- Establish a resource management information system with enabling support systems and services to maintain, track, allocate, and improve the resources for present and future organization needs. Computer-

based equipment tracking, facilities allocation, and other systems are recommended for organizations with over 50 people.

- Attend to physical factors, including facilities and human factors, such as ambient noise level and computer access to specific tools and applications.
- Begin planning in early life cycle stages of all system development efforts to address utilization and support resource requirements for system transition, facilities, infrastructure, information/data storage, and management. Enabling resources should also be identified and integrated into the organization's infrastructure.
- Engage project management, risk management, and business management processes to fully integrate Infrastructure Management processes to ensure organizational adoption.

Elaboration

Infrastructure Management Concepts. Projects all need resources to meet their objectives. Project planners determine the resources needed by the project and attempt to anticipate both current and future needs. The Infrastructure Management process provides the mechanisms whereby the organization infrastructure is made aware of project needs and the resources are scheduled to be in place when requested. While this can be simply stated, it is less simply executed. Conflicts must be negotiated and resolved, equipment must be obtained and sometimes repaired, buildings need to be refurbished, and information technology services are in a state of constant change. The infrastructure management organization collects the needs, negotiates to remove conflicts, and is responsible for providing the enabling organization infrastructure without which nothing else can be accomplished. Since resources are not free, their costs are also factored into investment decisions. Financial resources are addressed under the Portfolio Management process (see [Section 2.3.3.3](#)), but all other resources, except for human resources which are addressed under the Human Resource Management process (see [Section 2.3.3.4](#)), are addressed under this process.

Infrastructure management is complicated by the number of sources for requests, the need to balance the skills of the labor pool against the other infrastructure elements (e.g., computer-based tools), the need to maintain a balance between the budgets of individual projects and the cost of resources, the need to keep apprised of new or modified policies and procedures that might influence the skills inventory, and myriad unknowns.

Resources are allocated based on requests. Infrastructure management collects the needs of all the projects in the active portfolio and schedules or acquires nonhuman assets, as needed. Additionally, the infrastructure management process maintains and manages the facilities, hardware, and support tools required by the portfolio of organization projects.

Infrastructure management is the efficient and effective deployment of an organization's resources when and where they are needed. Such resources may include inventory, production resources, or information technology. The goal is to provide materials and services to a project when they are needed to keep the project on target and on budget. A balance should be found between efficiency and robustness. Infrastructure management relies heavily on forecasts into the future of the demand and supply of various resources.

The organization environment and subsequent investment decisions are built on the existing organization infrastructure, including facilities, equipment, personnel, and knowledge. Efficient use of these resources is achieved by exploiting opportunities to share enabling systems or to use a common system element on more than one project. These opportunities are enabled by good communications within the organization. Integration and interoperability of supporting systems, such as financial, human resources (see [Section 2.3.3.4](#)), and training, is critically important to executing organization strategic objectives. Feedback from active projects is used to refine and continuously improve the infrastructure.

Further, trends in the market may suggest changes in the supporting environment. Assessment of the availability and suitability of the

organization infrastructure and associated resources provides feedback for improvement and reward mechanisms. All organization processes require mandatory compliance with government and corporate laws and regulations. Decision making is governed by the organization strategic plan.

Infrastructure Management Process Maturity. The Infrastructure Management process primarily focuses on the establishment and deployment of infrastructure rather than the construction or actual use of the infrastructure. Since the quality of a product is related to the structure and use of the infrastructure employed, the maturity and quality of the process employed toward management of the infrastructure can help provide higher quality process inputs, outputs, and outcomes.

2.3.3.3 Portfolio Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.2.3.1] The purpose of the Portfolio Management process is to initiate and sustain necessary, sufficient, and suitable projects to meet the strategic objectives of the organization.

Portfolio management also provides organizational output regarding the set of projects, systems, and technical investments of the organization to external stakeholders, such as parent organizations, investors/funding sources, and governance bodies.

Description Projects create the products or services that meet the objectives and generate revenue for an organization. Thus, the conduct of successful projects requires an adequate allocation of funding and resources and the authority to deploy them to meet project objectives. Most business entities manage the commitment of financial resources using well-defined and closely monitored processes.